TITANIC
SURVIVAL PREDICTION

A supervised machine-learning approach to understanding passenger survival

PROJECT REPORT

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Prepared for: **MDI3003 - Advanced Predictive Analytics**

Dataset: Kaggle — Titanic: Machine Learning from Disaster

Date: 12th October 2026

Abstract

This project investigates whether passenger information can be used to predict survival in the sinking of the RMS Titanic. The task is framed as a binary classification problem: a passenger either survived (1) or did not survive (0). The Kaggle dataset provides a labelled training set of 891 passengers and an unlabelled test set of 418 passengers. The proposed workflow combines exploratory data analysis, missing-value treatment, feature engineering, categorical encoding, model training, validation, and generation of a Kaggle-compatible prediction file. Logistic Regression will provide an interpretable baseline, while tree-based models such as Random Forest will be evaluated for their ability to capture non-linear relationships. Final model selection will be based on validation performance and appropriate error analysis; no performance score is reported until experiments have been run.

1. Understanding of the Problem

The central question is: “What characteristics were associated with a greater likelihood of surviving the Titanic disaster?” The task uses historical passenger records to learn patterns from known outcomes and apply those patterns to passengers whose outcomes are withheld in the competition test set.

- Problem type: supervised binary classification.
- Target variable: Survived (0 = did not survive; 1 = survived).
- Training data: 891 records with survival labels.
- Test data: 418 records without survival labels; predictions are required for submission.
- Competition metric: accuracy, the proportion of predictions that match the hidden labels.
- Potential predictors: passenger class, sex, age, fare, family relationships aboard, embarkation port, and engineered features.

The model is intended to identify statistical patterns in this dataset, not to prove that any one feature caused survival. Historical context, incomplete records, and the limited size of the dataset should be considered when interpreting results.

2. Project Objectives

- Inspect and understand the structure, types, distributions, and quality of the passenger data.
- Identify missing values, unusual values, class imbalance, and potentially informative relationships.
- Prepare a repeatable preprocessing pipeline that avoids information leakage between training and validation data.
- Train and compare a simple baseline with more flexible classification algorithms.
- Evaluate performance using a validation split or cross-validation, including accuracy and a confusion matrix.
- Generate a submission file containing PassengerId and Survived for all 418 test records.
- Document limitations and identify future improvements.

3. Literature Review

3.1 Kaggle competition documentation

Kaggle's "Titanic: Machine Learning from Disaster" defines the task as predicting binary survival outcomes from passenger features. Its documentation describes the labelled training set, unlabelled test set, feature definitions, accuracy-based evaluation, and the required submission format. It also encourages feature engineering as part of the modelling process [1].

3.2 Sociological interpretation of survival patterns

Gupta, Sharma, and Bouza Herreras (2018) discuss Titanic survival as a problem shaped by social factors and use machine-learning methods to study variables including sex, title, and family density. Their work motivates analysing social and demographic features rather than treating the task as a purely numerical prediction exercise [2]. These findings are useful as hypotheses to test, not as results to assume for the current implementation.

3.3 Comparing classification algorithms

A recent comparative study by the authors listed in the 2026 EWA Publishing paper examines multiple machine-learning approaches for Titanic survival prediction and reports that tree-based approaches can provide useful predictive performance on this small classification dataset [3]. This supports comparing models empirically instead of choosing an algorithm solely from prior expectations.

3.4 Data preprocessing and categorical features

The scikit-learn documentation explains that missing data must be handled appropriately for estimators that cannot accept incomplete inputs, and that one-hot encoding converts categorical values into numerical indicator features for many machine-learning algorithms [4, 5]. The proposed workflow therefore places imputation and encoding inside a preprocessing pipeline fitted only on the training folds.

Literature review synthesis

The reviewed sources suggest three practical principles: (1) use demographic and socio-economic attributes as candidate predictors; (2) treat missing data and categorical values systematically; and (3) compare models using held-out validation rather than relying on a single algorithm or an unverified expected score. The current project will apply these principles and report its own results after evaluation.

4. Dataset Overview

Dataset / field	Description	Role
train.csv	891 passenger records; includes Survived	Model development and validation
test.csv	418 passenger records; target withheld	Final prediction
Survived	0 = no, 1 = yes	Target label
Pclass, Sex, Age	Class, sex, and age	Candidate predictors
SibSp, Parch	Family relationships aboard	Candidate predictors / feature engineering
Fare, Embarked, Cabin, Ticket, Name	Fare, port, cabin, ticket, and name fields	Candidate predictors / derived features

The training file contains 12 columns including the target; the test file contains the corresponding passenger features without the target. Some fields, particularly Age, Cabin, and Embarked, contain

missing values. Cabin and Ticket may be sparse or high-cardinality, so their use will be evaluated carefully.

5. Proposed Methodology and Plan

Step 1 — Data loading and audit: Load both CSV files; inspect dimensions, column names, data types, duplicate records, target balance, and missing-value counts.

Step 2 — Exploratory data analysis: Visualise survival by sex, passenger class, age group, fare range, family size, and embarkation port. Use plots to form hypotheses, not to claim causation.

Step 3 — Preprocessing: Impute numerical values (for example, median Age/Fare) and categorical values (for example, most frequent category); encode categorical predictors.

Step 4 — Feature engineering: Test features such as FamilySize = SibSp + Parch + 1, IsAlone, extracted title from Name, and possibly fare per person. Keep only features justified by validation.

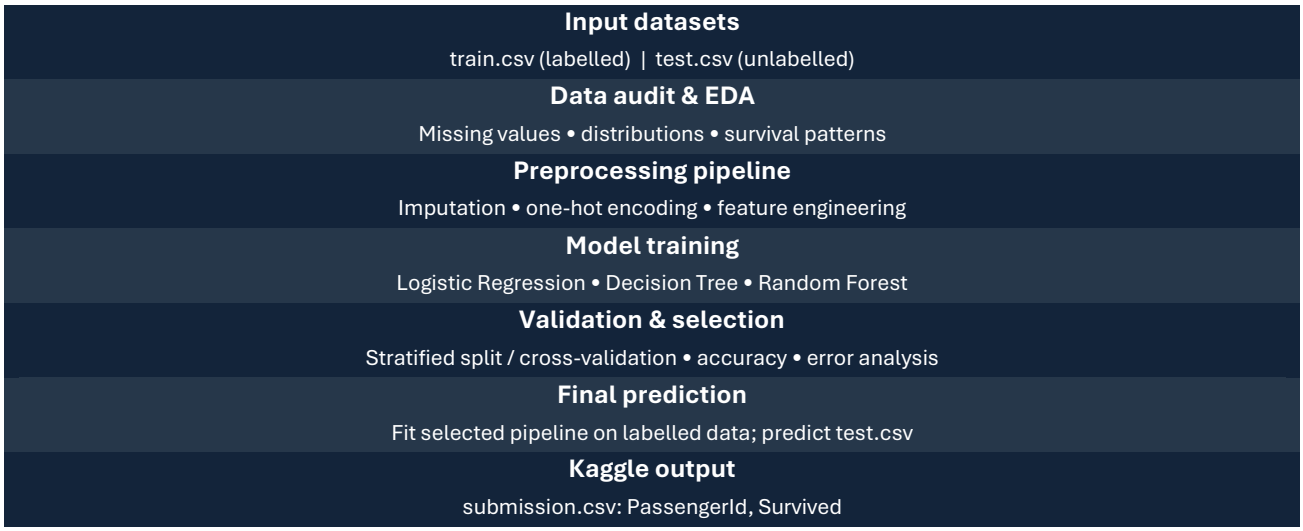
Step 5 — Baseline model: Establish a simple baseline and train Logistic Regression with preprocessing in a single pipeline.

Step 6 — Model comparison: Evaluate Decision Tree and Random Forest; consider a gradient-boosting model if time and tooling permit.

Step 7 — Validation and tuning: Use a stratified train/validation split or stratified cross-validation. Compare accuracy and inspect precision, recall, F1-score, and confusion matrix.

Step 8 — Final training and submission: Select the best validated pipeline, fit it on all labelled training data, predict the 418 test records, and export PassengerId, Survived.

6. Proposed Model Architecture / Workflow



7. Work Done Till Now / Current Development Status

The project's modelling workflow has been defined, including the data preparation strategy, candidate predictors, baseline model, model-comparison approach, validation plan, and Kaggle submission format. The current development phase is focused on preparing the dataset and organising the training-

and-validation pipeline. Quantitative model performance, feature importance, and the final submission score will be recorded after the experiments are executed and checked.

Workstream	Status / next action
Problem definition and dataset review	Defined
Preprocessing and feature engineering design	Planned and being prepared
Model training and validation	Development phase; performance to be measured
Kaggle submission and leaderboard evaluation	Pending final model selection and prediction export

8. Evaluation Strategy

- Primary metric: accuracy, matching the competition's stated evaluation metric.
- Supporting metrics: precision, recall, F1-score, and confusion matrix to understand types of errors.
- Validation: stratified hold-out split or stratified cross-validation to preserve target proportions.
- Leakage prevention: fit imputers, encoders, and feature transformations on each training fold only.
- Model selection: compare validation results and consistency, not training accuracy alone.

No numerical accuracy or leaderboard result is included at this stage because the model experiments have not yet been verified. The report should be updated with measured values once training and validation are completed.

9. Expected Outcomes

- A reproducible machine-learning pipeline for Titanic survival classification.
- A comparison of baseline and tree-based classifiers using a consistent validation method.
- A clearer understanding of which available features are associated with survival predictions.
- A Kaggle-formatted prediction file with 418 rows and the required two columns.
- A documented discussion of limitations, including small sample size, missing data, historical bias, and uncertainty in predictions.

10. Limitations and Ethical Considerations

The dataset is small, historical, and incomplete. Some demographic and socio-economic variables may act as proxies for unequal access to lifeboats or evacuation opportunities. A model can learn associations in the recorded data but cannot reconstruct every decision or circumstance during the disaster. Results should be interpreted as a learning exercise in predictive modelling, not as a definitive causal explanation of survival.

11. Conclusion

This project establishes a structured approach to predicting Titanic passenger survival using supervised machine learning. The plan combines data inspection, principled preprocessing, feature engineering, baseline modelling, tree-based model comparison, validation, and submission generation. The next milestone is to execute the training pipeline, record validation metrics, select a model based on evidence, and prepare the final Kaggle submission. The completed analysis will report actual measured results rather than assumed performance.

References

[1] Kaggle. “Titanic — Machine Learning from Disaster.” Competition overview and data documentation. https://www.kaggle.com/c/titanic

[2] Gupta, K., Sharma, P., & Bouza Herreras, C. N. (2018). “Surviving the Titanic tragedy: A sociological study using machine learning models.” Suma de Negocios. https://www.encyclopedia-titanica.org/surviving-the-titanic-tragedy-a-sociological-study-using-machine-learning-models.html

[3] “Survival Prediction of the Titanic: A Comparison of Multiple Machine Learning Algorithms.” (2026). EWA Publishing. https://doi.org/10.54254/2755-2721/2026.34283

[4] scikit-learn. “Imputation of missing values.” User Guide. https://scikit-learn.org/stable/modules/impute.html

[5] scikit-learn. “OneHotEncoder.” API documentation. https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html